

SHIELD-MOEA: Scenario Screening with Disjoint Evaluation for Distribution-Network Resilience Planning

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Abstract

Distribution network planning under load and outage uncertainty requires portfolios that remain useful as binding stresses change. SHIELD-MOEA extends non-dominated sorting with selective worst-scenario screening and disjoint scenario draws for search and final evaluation. On a SimBench-derived benchmark with 72 candidate actions, five objectives, and eight archived experiment labels, the method reaches pooled mean hypervolume 0.2740 over 30 seeded runs per stochastic method. The eight labels represent five active proxy configurations because the configured DER-output multiplier does not enter any p4 optimization or scoring objective; the `der_uncertainty` block is therefore an inactive-factor replicate, not DER-uncertainty evidence. The pooled result is 5.09% above NSGA-II with post-hoc final-population repair and 5.56% above plain NSGA-II. All 32 comparisons with four stochastic baselines are Holm-significant wins; sixteen gaps to two deterministic rules are descriptive and favor SHIELD-MOEA. The margin persists for sampled worst-envelope hypervolume (+5.36%) and under held-out load/outage ranges (+6.04%). Feasibility repair carries the main measured gain. Static code-path accounting gives 65% fewer search-phase plan-scenario objective rows with screening, although no front-quality difference is detected against no screening; hybrid variation and periodic re-screening are also unresolved against simpler controls. A 1296-case composition-level AC check across six networks yields aggregate feasibility 0.685, compared with 0.389 without planning and 0.694 for post-hoc-repaired NSGA-II. The evidence supports a disjoint-evaluation workflow under the implemented load/outage proxies while identifying repair-rather than every named operator-as the resolved performance mechanism.

Keywords: distribution network planning; power system resilience; scenario screening; multi-objective evolutionary optimization; DER integration actions; NSGA-II; SimBench; AC power flow validation

1. Introduction

Resilience has moved from the margins of distribution planning to its center. Storm-driven outages, heat-wave load peaks, rooftop photovoltaics, and other distributed energy resources (DERs) create stress combinations not represented by historical load-duration curves. Reliability-report archives document the resulting operational and economic consequences [1,2]. At the same time, feeder reinforcement, storage, DER integration, and automation interact through a shared budget, and their value depends on the realized stress scenario. The planning decision is therefore inherently multi-objective: investment cost against losses, voltage exposure, reliability, and survivability under outages, all under scenario uncertainty.

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Current methods leave a gap at the interface between the optimizer and the uncertainty model. Robust planning formulations can focus on a single worst case and scalarize the objective space, limiting visibility of the cost-resilience trade surface [3,4]. Evolutionary multi-objective methods expose that surface, but commonly evaluate candidates on a scenario set fixed before search [5,6]. The scenarios binding for mature plans, however, may differ from those affecting the initial population. When the same scenario sample steers and grades the search, sample-specific adaptation is also difficult to distinguish from robustness. Distribution utility planning departments instead require investment fronts that remain defensible under stress scenarios not used to tune the plans.

This paper proposes SHIELD-MOEA, a scenario-selective extension of non-dominated sorting for distribution resilience planning. The algorithm retains a standard multi-objective backbone—crowding-based selection, hybrid GA/DE variation, and greedy feasibility repair—and places its contribution at the optimizer–uncertainty interface. Every few generations, it identifies the K scenarios on which the current population performs worst and uses them for search-phase fitness. The final front is evaluated on a complete scenario set generated with disjoint random seeds, preventing direct reuse of search realizations in the reported score. We report mean and sampled worst-envelope hypervolume and supplement the proxy evaluation with pandapower AC power flow.

Real resilience-investment records are utility-internal, so evaluation uses a SimBench-derived proxy benchmark [7]. Statistics from the 18 highest-ranked subnets generate 72 candidate actions. The archive has eight experiment labels but only five active proxy configurations: four labels share the reference objective configuration under different optimizer seed streams, including `der_uncertainty`, because the configured DER-output multiplier is inactive in the p4 objective vector. The unseen-stress setting changes held-out load and outage ranges that do enter the proxy; its changed DER range does not. Shared infrastructure with CARS-MODE is declared in Section 2.4 and the Data Availability Statement.

The contributions of this paper are:

1. **A disjoint-evaluation protocol for scenario-aware evolutionary planning.** SHIELD-MOEA scores final fronts on scenario draws not used during search, preventing direct reuse of the realizations that guided optimization. An unseen-range experiment provides an additional test of search-range specialization (Sections 4.3 and 4.6).
2. **A selective worst- K screening mechanism with quantified code-path economy.** At the default configuration, static accounting of evaluated objective-array rows falls by 65% (51,200 to 17,920). No front-quality difference is detected against no screening, and periodic re-screening remains unresolved against a fixed generation-1 subset; screening is therefore an exposure-and-call-count mechanism rather than a demonstrated quality or wall-clock mechanism (Section 6.3).
3. **A statistically grounded framework result with bounded component attribution.** Across eight archived labels and 30 seeds per stochastic method, SHIELD-MOEA reaches pooled mean hypervolume 0.2740, +5.09% over post-hoc-repaired NSGA-II and +5.56% over plain NSGA-II, with 32/32 Holm-significant stochastic-baseline wins. The pooled summary mixes five active configurations and four reference-configuration seed blocks. Within the implemented SHIELD stack, repair is the dominant resolved component (-8.10% when removed), whereas hybrid variation, dynamic re-screening, and removing survivability from environmental selection remain unresolved against their controls (Sections 6.1 and 6.3).
4. **Sampled-envelope, sensitivity, and cross-family AC diagnostics.** The method retains +5.36% sampled worst-envelope HV and +6.04% when held-out load and outage ranges move beyond the search ranges; the simultaneously changed DER range is

inactive in the p4 proxy. A 1296-case pandapower check across six networks yields aggregate feasibility 0.685 versus 0.389 without planning, while post-hoc-repaired NSGA-II reaches 0.694. A one-at-a-time parameter sweep reports the setting at which separation from NSGA-II is lost ($N_p = 60$, $p = 0.104$) rather than presenting only favorable points (Sections 6.2, 6.4, and 6.5).

Screening concentrates search-phase effort on scenarios difficult under the implemented load/outage proxy. At the default setting, static code-path accounting is 65% lower without a detected change in mean front quality; the screening-off ablation is 0.48% lower, and periodic re-screening does not outperform a generation-1 fixed worst- K set. Because the objectives are closed-form and no runtime counter records the analytic call total, this arithmetic does not establish a wall-clock saving. Screening therefore supplies selective exposure and a computed objective-row reduction, while repair carries the resolved component gain.

The paper proceeds as follows. Section 2 reviews related work. Section 3 defines the planning problem and the public benchmark. Section 4 describes SHIELD-MOEA. Section 5 gives the experimental protocol. Section 6 reports results, Section 7 discusses them, Section 8 states limitations, and Section 9 concludes.

2. Related Work

SHIELD-MOEA draws on three bodies of literature that have so far evolved largely in parallel: resilience-oriented distribution planning under $N-k$ and extreme-weather uncertainty, scenario generation and reduction for stochastic power system optimization, and hybrid multi-objective evolutionary algorithms with robustness-aware objectives. Below we discuss each thread and identify the gap that motivates this paper.

2.1. Resilience-Oriented Distribution Planning and Hardening Decisions

Panteli and Mancarella [8] distinguish high-impact, low-probability events from ordinary reliability and hardening from operational adaptation. Bhusal et al. [9] document the rapid growth of planning-stage resilience research. The dominant paradigm is robust optimization over an $N - k$ or weather-driven uncertainty set. Yuan et al. [3] formulate a two-stage robust problem with adversarial component outages. Ma et al. [5] extend this logic to joint hardening and distributed-generation placement with wind-dependent fragility, while Lin and Bie [6] add reconfiguration and DG islanding. Panteli et al. [10] connect weather intensity to component failure through fragility curves.

Practice-oriented studies consider resilient storage siting [11], zone-wise planning under decision-dependent uncertainty [12], and genetic-algorithm-driven feeder hardening [13]. Related work addresses resilient renewable microgrids [14] and islanding with fault recovery [15]. A parallel strand grounds planning claims in full AC physics. Mahmoudi and Alizadeh quantify errors caused by linearized expansion-planning models [16], while Trivic and Savic combine NSGA-II storage planning with power-flow analysis [17]. This evidence motivates the AC validation layer in Section 6.5.

A consistent pattern emerges across these studies. The trilevel robust models protect against a single worst case and collapse the objective space to one scalar, so the planner never sees the cost-resilience trade surface. The metaheuristic studies expose multiple objectives, but they evaluate candidate plans on a scenario set fixed before the search starts, and the same scenarios serve both to steer the search and to grade its output. None of them guards against a plan that has quietly overfitted the scenario sample it was optimized on.

2.2. Scenario Generation, Reduction, and Screening in Stochastic Power System Optimization

The second thread represents uncertainty through scenarios. Classical reduction selects a subset while minimizing probability distance; reference approaches include the Fortet–Mourier formulation [18] and forward/backward reduction [19]. Power-system applications adopted these methods early [20], and importance sampling made stochastic unit commitment tractable [21]. Generative models later reproduced spatiotemporal renewable behavior without parametric assumptions [22]. Applications include stochastic microgrid dispatch [23], distribution reconfiguration [24], and joint siting and sizing [25].

Recent work moves scenario machinery inside optimization. Li et al. learn an ambiguity set for distributionally robust DG planning [26], while Zhou et al. use extreme-event scenarios in integrated energy planning [27]. Decision-dependent uncertainty [28] is the closest mathematical-programming analogue to a scenario set that changes with candidate decisions. Wu et al. preserve extreme wind, PV, and load combinations during generation [29], paralleling the worst- K focus used here.

Most reductions are executed once, judged by distributional fidelity, and reused for every evaluation. The scenarios guiding search are therefore also used to grade the result. This distinction may be less important in a single monolithic solve. In population search, however, thousands of plans are compared while the binding scenarios can change with the population. SHIELD-MOEA separates search guidance from final scoring and tests whether updating the guidance subset adds value.

2.3. Hybrid Multi-Objective Evolutionary Algorithms and Robustness-Aware Objectives

The third thread concerns the optimizer. NSGA-II [30] and MOEA/D [31] remain common frameworks, with hypervolume [32] measuring front quality. Differential evolution [33] supplies mutation behavior complementary to GA crossover, and Das and Suganthan [34] review blended DE strategies. Jin and Branke [35] survey optimization under uncertain fitness, while Deb and Gupta [36] treat robustness as an objective over perturbation neighborhoods.

Recent work evaluates candidate solutions across fixed scenario sets [37] or combines decision preferences with variable-level uncertainty [38]. In both cases, the uncertainty set remains fixed during the run. SHIELD-MOEA tests periodic scenario-subset updates and, through a fixed worst- K control, whether those updates matter.

Recent power-planning studies hybridize evolutionary variation. Examples include an improved NSGA-II for storage-and-EV co-optimization [39], a three-stage TER-NSGA-II for resilient backbone-grid planning [40], and an enhanced beluga whale optimizer for storage planning [41]. Under uncertainty, such methods commonly average fitness over a fixed scenario sample or solve a robust reformulation before evolutionary search. The former incurs repeated evaluation cost, whereas the latter leaves the scenario set outside the evolving search state.

SHIELD-MOEA instead treats the evaluated scenario subset as a search-time quantity. The subset is periodically updated toward scenarios that are difficult for the current population. Evaluation on a disjoint scoring set then reports both mean and sampled worst-case hypervolume. The targeted controls in Section 6.3 test whether periodic updating adds value beyond a fixed generation-1 worst- K set.

2.4. Relation to CARS-MODE and Gap Statement

CARS-MODE [42] also studies evolutionary distribution planning on public benchmark-derived portfolios. It adapts DE control parameters and repairs budget violations in an economic DER-and-storage setting. SHIELD-MOEA instead studies the optimizer–uncertainty interface. Its NSGA-II-style kernel combines GA/DE variation and repair,

while screening supplies search scenarios that remain disjoint from final scoring draws. The current p4 proxy responds to load and outage coordinates; its configured DER-output coordinate is inactive, although DER-oriented actions remain in the candidate pool and DER stress is present in the separate AC layer. The studies share the SimBench extraction, 72-action builder, binary-plan and budget representation, evaluation/statistics utilities, seeded archive machinery, and AC composition-mapping infrastructure. Their research questions, method-specific search configurations, run archives, and conclusions remain separate.

The gap lies at the interface of search and uncertainty. Resilience studies often grade a robust solution or Pareto front on the same scenarios that guide optimization. Offline reduction is blind to the evolving population, while hybrid MOEAs usually adapt operators rather than exposure. SHIELD-MOEA combines worst-scenario screening, disjoint-draw final scoring, and sampled worst-case hypervolume reporting. The fixed worst- K control determines whether the adaptive update, rather than screening alone, is necessary.

3. Problem Formulation and Public Benchmark

3.1. Resilience-Oriented Planning as Scenario-Aware Multi-Objective Selection

Let $\mathcal{A} = \{a_1, \dots, a_n\}$ be the candidate-action pool. Action a_i has synthetic cost c_i and coefficients $(l_i, u_i, h_i, r_i, g_i, d_i)$ for loss reduction, voltage-risk reduction, hosting-capacity gain, reliability gain, survivability gain, and a DER-readiness diagnostic. These coefficients are derived from public network statistics as described in Section 3.2. A plan is a binary vector $x \in \{0, 1\}^n$. The configured scenario is $s = (\lambda, \delta, \sigma)$, where λ is a load multiplier, δ is a DER-output multiplier, and $\sigma \in [0, 1]$ is outage severity.

For plan x under scenario s , the benchmark defines

$$\begin{aligned} L(x, s) &= \max\left(0.015, L_0\lambda - \frac{1}{120} \sum_{i=1}^n l_i x_i\right), \\ U(x, s) &= \max\left(0.005, U_0\lambda - \frac{1}{10} \sum_{i=1}^n u_i x_i\right), \\ H(x, s) &= \min\left(1, \frac{\sum_{i=1}^n h_i x_i}{\max(1, H_0\delta)}\right), \\ R(x) &= \min\left(1, 0.35 + \frac{1}{28} \sum_{i=1}^n r_i x_i\right), \\ S(x, s) &= \min\left(1, 0.42(1 - \sigma) + \frac{1}{24} \sum_{i=1}^n g_i x_i\right). \end{aligned}$$

Here, for total selected-source load P_Σ and line length D_Σ , the implementation sets $L_0 = 0.12 + 0.015D_\Sigma / \max(1, P_\Sigma)$, $U_0 = 0.18 + 0.010P_\Sigma / \max(1, D_\Sigma)$, and $H_0 = 0.45P_\Sigma$. L is a loss index, U is a voltage-risk index, H is a hosting diagnostic, R is a scenario-invariant reliability proxy, and S is a survivability proxy driven only by outage severity and selected actions. The experiment labeled `restoration_aware_evaluation` replaces L by L^ℓ during both search and evaluation:

$$L^\ell(x, s) = L(x, s) [1 + 0.30\sigma(1 - S(x, s))],$$

which couples loss exposure to plan survivability. With investment cost $C(x) = \sum_{i=1}^n c_i x_i$ and scenario set $\mathcal{S}$, the implemented p4 five-objective problem is

$$\begin{aligned} \min_{x \in \{0,1\}^n} \quad & F(x) = \left(C(x), \mathbb{E}_{s \in \mathcal{S}}[L(x, s)], \mathbb{E}_{s \in \mathcal{S}}[U(x, s)], \right. \\ & \left. - R(x), -\mathbb{E}_{s \in \mathcal{S}}[S(x, s)] \right) \\ \text{subject to} \quad & C(x) \leq B. \end{aligned} \quad 214$$

The budget B is the only hard constraint, and violation is measured as relative budget overrun. The p4 objective vector omits H , so δ does not affect p4 search objectives, held-out objectives, hypervolume, or compromise choice. The coefficient h_i is nevertheless implemented in H and enters the repair and Weighted Sum heuristics specified below; it is not an advertised p4 objective. DER-output variation becomes active only in the separate AC operating scenarios. Accordingly, L , U , R , and S are transparent engineering proxies rather than power-flow solutions, and the `der_uncertainty` planning label is not evidence of DER-output robustness. 215 216 217 218 219 220 221 222

3.2. Candidate Pool Derived from SimBench 223

Candidate attributes derive deterministically from the SimBench complete mixed dataset [7]. Eligible subnets are ranked by $p_j + 0.2\ell_j$, where p_j is total active load and ℓ_j is total line length; the highest 18 each supply reinforcement, storage, DER-integration, and automation actions. Reactive load and the extracted average loading field are not used by the candidate builder. Let m_j be line count, n_j load count, e_j renewable capacity, $\tau_j = p_j / \max(0.2, \ell_j)$, and $q_j = \max(0, 0.55p_j - e_j)$. In the attribute order $(c_i, l_i, u_i, h_i, r_i, g_i, d_i)$, the four exact implemented constructions are 224 225 226 227 228 229 230

$$\begin{aligned} \text{reinforcement: } & (60 + 4.5\ell_j + 7p_j, 0.012\ell_j + 0.020\tau_j, 0.020 + 0.006\tau_j, 0.06q_j, 0.018m_j, 0.016m_j, 0.20), \\ \text{storage: } & (50 + 16\sqrt{p_j + 1}, 0.025\sqrt{p_j + 1}, 0.018\sqrt{\tau_j + 1}, 0.12q_j + 0.08p_j, 0.055\sqrt{n_j + 1}, 0.070\sqrt{n_j + 1}, 0.20), \\ \text{DER integration: } & (45 + 10\sqrt{p_j + 1}, 0.018\sqrt{p_j + 1}, 0.012, 0.18q_j + 0.10p_j, 0.020\sqrt{n_j + 1}, 0.028\sqrt{n_j + 1}, 0.20), \\ \text{automation: } & (38 + 1.8m_j, 0.006m_j, 0.010\sqrt{\tau_j + 1}, 0.025q_j, 0.085\sqrt{m_j + 1}, 0.115\sqrt{m_j + 1}, 0.35). \end{aligned}$$

The resulting pool contains $n = 72$ candidates under nominal budget $B = 920$. Every attribute traces to the listed public-data fields through these rules, but the pool is a proxy rather than a costed engineering catalog and its cost unit is synthetic (Section 8). 232 233 234

3.3. Scenario Model and the Eight Archive Labels 235

Each label fixes ranges for the three configured scenario coordinates and draws $|\mathcal{S}| = 16$ scenarios uniformly. The search draw uses NumPy seed 20260713 and the evaluation draw seed 20260714; these two matrices are fixed and common to all methods and optimizer runs under a label. Table 1 lists the eight archived labels and distinguishes configured changes from changes that actually enter the p4 objective vector. 236 237 238 239 240

Thus the eight labels represent five active proxy configurations: the reference configuration is repeated under four labels with method- and label-specific optimizer seeds, and four additional labels activate load widening, outage widening, the L^e objective, or held-out load/outage ranges. The last label tests distribution shift in the two active scenario coordinates; its non-overlapping δ range adds no p4 planning stress. 241 242 243 244 245

3.4. Shared Public Benchmark Statement 246

This paper and CARS-MODE [42] use a shared planning runner, SimBench subnet extraction and 72-action builder, binary-plan representation, budget handling utilities, fixed-bound hypervolume helpers, seeded archive/statistics machinery, and AC composition 247 248 249

Table 1. The eight archive labels. Ranges are uniform sampling intervals for (load multiplier lambda, DER multiplier delta, outage severity sigma).

Experiment label	lambda range	delta range	sigma range	Implemented p4 effect
deterministic_vs_scenario	0.95-1.25	0.7-1.3	0.00-0.25	reference proxy configuration
der_uncertainty	0.95-1.25	0.4-1.7	0.00-0.25	inactive-factor replicate: δ is not consumed by p4 objectives
load_uncertainty	0.85-1.45	0.7-1.3	0.00-0.25	widened load band
outage_contingency	0.95-1.25	0.7-1.3	0.10-0.55	elevated outage severity
restoration_aware_evaluation	0.95-1.25	0.7-1.3	0.10-0.55	L^e replaces L in search and evaluation
scenario_screening_efficiency	0.95-1.25	0.7-1.3	0.00-0.25	reference proxy configuration under a separate optimizer
pareto_quality	0.95-1.25	0.7-1.3	0.00-0.25	seed stream
unseen_stress_generalization	0.95-1.25 (search)	0.7-1.3 (search)	0.00-0.25 (search)	reference proxy configuration under a separate optimizer seed stream evaluation uses $\lambda \in [1.3, 1.6]$, $\delta \in [1.4, 1.9]$, $\sigma \in [0.4, 0.7]$; only the λ and σ shifts affect p4 scores

mapper. Four proxy coordinates are common; CARS-MODE uses hosting capacity as its fifth objective, whereas p4 uses survivability. The paper-specific assets are the research question, method branch and controls, configurations, run archives, comparisons, and conclusions. Shared code and evaluation infrastructure do not make either paper’s results evidence for the other.

4. SHIELD-MOEA

Figure 1 closes the method’s scenario-search loop. Each population induces a worst-K screened scenario subset, the hybrid variation operators generate candidate plans, budget repair restores feasibility, and robust environmental selection returns survivors. Unseen-stress and AC evaluations branch from the final archive and remain outside the optimization loop.

SHIELD-MOEA exposes screening, repair, outage information, and survivability in environmental selection as individual switches in the main ablation study. Additional within-SHIELD controls compare hybrid variation with GA-only and DE-only offspring and periodic re-screening with a fixed generation-1 worst-K subset. These controls support only the operator scopes stated in Sections 4.2–4.5. The method is a resilience-oriented extension of NSGA-II [30]: it retains constraint-dominated non-dominated sorting and crowding-distance selection, and adds hybrid variation, scenario screening, and feasibility repair.

Formal definitions. For binary plan x , cost feasibility is

$$C(x) = \sum_{j=1}^n c_j x_j, \quad v_B(x) = \frac{\max(0, C(x) - B)}{B}.$$

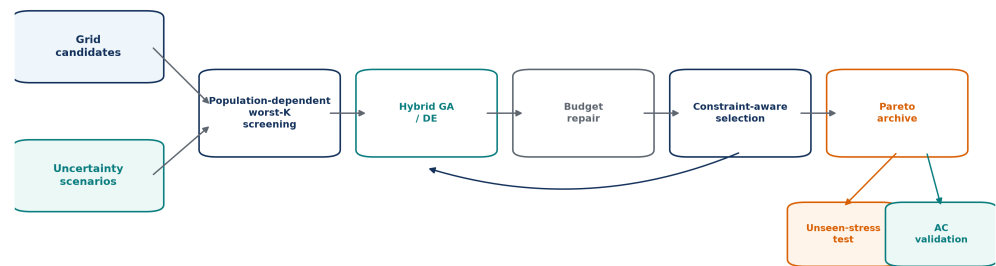


Figure 1. SHIELD-MOEA architecture. Scenario screening is population dependent; the feedback arrow denotes generational updating. The two validation branches use held-out stress scenarios or AC power flow and do not feed into selection.

Let l_q^{search} and u_q^{search} be fixed bounds generated on the search problem from the reference-plan sample in Section 5.3. At a screening generation, the implemented population-level scenario score is

$$\rho_t(s) = \frac{1}{|P_t|} \sum_{x \in P_t} \left[\sum_{q=1}^5 \frac{f_q(x; s) - l_q^{\text{search}}}{u_q^{\text{search}} - l_q^{\text{search}}} + 10v_B(x) \right].$$

The active subset is the worst K scenarios,

$$\mathcal{S}_t^K = \arg \max_{\substack{\mathcal{A} \subseteq \mathcal{S}_{\text{search}} \\ |\mathcal{A}|=K}} \sum_{s \in \mathcal{A}} \rho_t(s).$$

These screening normalizations are not clipped and use equal unit weights; cost and reliability add scenario-constant offsets, while λ and σ determine the scenario ordering and δ has no p4 effect. Between screening generations the active set is held fixed. Environmental selection uses the objective average over that set,

$$\bar{f}_q^K(x; t) = \frac{1}{K} \sum_{s \in \mathcal{S}_t^K} f_q(x; s).$$

The GA channel applies uniform crossover followed by independent bit mutation,

$$o_j^{GA} = \begin{cases} x_j^{(a)}, & U_j < 0.5, \\ x_j^{(b)}, & \text{otherwise,} \end{cases} \quad o_j^{GA} \leftarrow 1 - o_j^{GA} \text{ with probability } 1/n.$$

For the DE channel, a relaxed donor is formed and mapped to a stochastic bit,

$$z = x_{r_1} + 0.5(x_{r_2} - x_{r_3}), \quad o_j^{DE} \sim \text{Bernoulli}(0.08 + 0.82 \mathbb{I}[z_j > 0.5]).$$

Repair uses the code's fixed, p4-independent candidate score

$$b_j = \frac{l_j}{0.12} + \frac{u_j}{0.02} + \frac{h_j}{5} + \frac{r_j}{2} + \frac{g_j}{2}, \quad j^- = \arg \min_{j: x_j=1} \frac{b_j}{c_j}, \quad x_{j^-} \leftarrow 0,$$

repeating until $v_B(x) = 0$. Held-out evaluation forms both mean and componentwise-worst objective vectors,

$$f_q^{\text{mean}}(x) = \frac{1}{|\mathcal{S}_{\text{eval}}|} \sum_{s \in \mathcal{S}_{\text{eval}}} f_q(x; s), \quad f_q^{\text{worst}}(x) = \max_{s \in \mathcal{S}_{\text{eval}}} f_q(x; s).$$

The mean-front plans are evaluated under both aggregations. Hypervolume alone applies clipped fixed-bound normalization, separately for mean and worst aggregations, followed by the standard dominated-volume definition,

$$HV_z(\mathcal{P}; r_z) = \lambda_Q \left(\bigcup_{x \in \mathcal{P}} \prod_{q=1}^Q [\tilde{f}_{q,z}(x), r_{q,z}] \right), \quad z \in \{\text{mean, worst}\}.$$

Under the implemented schedule, static counting of objective-array rows implied by the code path gives

$$C_{\text{screened}} = GN_u K + N_r N_p |\mathcal{S}| = 12,800 + 5,120 = 17,920,$$

where union evaluation and ranking-round evaluation are disjoint terms. The implementation does not maintain a runtime call counter, so this is reproducible code-path arithmetic rather than an instrumented measurement or wall-clock claim.

4.1. Non-Dominated Sorting Kernel (NSGA-II Framework)

The kernel maintains a population of 40 binary plans over 40 generations. Environmental selection is constraint-dominated non-dominated sorting (feasible plans dominate infeasible ones; infeasible plans compare by relative budget violation) with crowding-distance truncation, following the standard NSGA-II procedure [30]. Initial plans are sparse random selections (3-18% density) and are repaired before generation 1 when repair is enabled. The baseline operator and repair-timing differences are stated in Section 5.1, so a framework-level margin is not attributed to the scenario interface alone.

4.2. Hybrid GA/DE Variation

Each generation produces 40 offspring. Half use uniform crossover followed by bit-flip mutation at rate $1/n$. The other half use the displayed DE-style relaxed donor followed by stochastic bit sampling with probabilities 0.9 and 0.08 above and below the strict 0.5 threshold. The three donor indices are sampled independently with replacement, so this channel is not a canonical distinct-index DE/rand/1 implementation. Section 6.3 compares this hybrid with controls that select GA-only or DE-only offspring while otherwise retaining the same SHIELD code path.

4.3. Worst-K Scenario Screening (The Adaptive Interface)

At generation 1 and every $T_s = 5$ generations thereafter, the algorithm evaluates the current population on each of 16 search scenarios, ranks scenarios by $\rho_t(s)$ (worst first), and retains the top $K = 4$. Until the next screening event, parent-offspring selection uses only the active set. Let $N_p = 40$, $N_u = 2N_p = 80$, $G = 40$, $|\mathcal{S}| = 16$, and $N_r = 8$ screening rounds (generations 1, 6, ..., 36). The unscreened path implies $C_{\text{full}} = GN_u |\mathcal{S}| = 51,200$ objective rows. The screened path implies disjoint terms $C_{\text{active}} = GN_u K = 12,800$ for union selection and $C_{\text{rank}} = N_r N_p |\mathcal{S}| = 5,120$ for ranking the current population, totaling 17,920. Thus the static path count is 65% lower; wall-clock behavior is evaluated separately.

4.4. Feasibility Repair

For SHIELD-MOEA, every initial and offspring plan exceeding the budget is repaired greedily with b_j/c_j before environmental selection. The NoRepair ablation skips both timings. The comparator labeled NSGA-II+Repair is different: it runs constrained NSGA-II without repair during search and applies the same drop rule only to the returned population. The ablation therefore identifies the implemented SHIELD repair schedule, while the NSGA-II comparison is rule-matched only at post-processing, not timing-matched.

4.5. Resilience Objective and Outage Awareness in Search

The survivability objective $-S(x, s)$ is part of environmental selection by default. Ablation-NoResilienceObj drops this column from parent-offspring environmental selection but, as implemented, the worst- K screening score ρ_t still uses all five objectives. Its result therefore tests removal from environmental selection, not complete removal of survivability information from search. Ablation-NoOutage instead sets $\sigma = 0$ throughout search, affecting both screening and environmental selection while leaving evaluation unchanged. The scope of each contrast is retained in Sections 6.3 and 6.5.

4.6. Direct-Reuse-Leakage-Controlled Evaluation and Worst-Envelope Rationale

No method—proposed, baseline, or ablation—is scored on the fixed scenarios it searched on. The evaluation scenario matrix is drawn with seed 20260714 rather than the search seed 20260713. In the unseen-stress label, the active load and outage ranges are also disjoint; the changed DER range is inert in the proxy. The final population is re-evaluated on this held-out matrix, filtered for budget feasibility, and reduced to a non-dominated front in raw objective space before any hypervolume normalization. Screening influences which plans are found but has no access to the held-out matrix.

We report two hypervolume variants. Standard mean hypervolume averages each objective over the 16 evaluation scenarios. The sampled worst-envelope readout reuses the plans on the mean-objective non-dominated front and replaces each scenario-dependent objective by its componentwise maximum over those 16 scenarios before hypervolume calculation; it does not construct or optimize a separate worst-objective front. This is a sample-dependent diagnostic rather than a bound on tail behavior. The two metrics use separate fixed normalization bounds (Section 5.3) and are comparable across methods only within their respective columns.

4.7. Algorithm Summary

```
SHIELD-MOEA(candidates, budget, search_scenarios, seed):
    initialize 40 sparse random plans; repair each to budget      # 4.4
    active <- search_scenarios
    for gen = 1 .. 40:
        if screening enabled and gen = 1 (mod 5):                # 4.3
            score each search scenario by population-mean scalarized fitness
            active <- worst 4 scenarios
        offspring <- 20 by uniform crossover + bit-flip (1/n)    # 4.2
            + 20 by the implemented DE-style stochastic donor
        repair offspring to budget                                # 4.4
        F <- five-objective values of parents+offspring on active # 4.5
        population <- constraint-dominated NDS + crowding (top 40) # 4.1
    return final population
wrapper re-evaluates it on the DISJOINT evaluation scenarios,
    filters budget-feasible plans, and extracts the raw-objective front
```

5. Experimental Setup

5.1. Methods Compared

Table 2 lists eleven methods: SHIELD-MOEA, six baselines, and four single-switch ablations. Pymoo supplies NSGA-II and MOEA/D [43]. All methods see the same binary problem and fixed search-scenario matrix within a label, but their operators, constraint handling, returned object, repair timing, and—for MOEA/D—population size are not identical.

Table 2. Methods compared.

Method	Role	Description
SHIELD-MOEA	proposed	NSGA-II-style kernel + GA/DE hybrid variation + worst-K scenario screening + repair
NSGA-II	baseline	pymoo NSGA-II, population 40, sparse Boolean sampling, two-point crossover, bit-flip mutation, budget constraint, all 16 search scenarios
NSGA-II+Repair	baseline	the same constrained NSGA-II, followed by SHIELD's drop rule on the returned population only; no repair during search
MOEA/D	baseline	pymoo MOEA/D with 35 Das–Dennis reference directions, two-point crossover, bit-flip mutation, and $10^4 v_B$ objective penalty
GA	baseline	single returned point from tournament/uniform-crossover GA with mutation $1.5/n$, unclipped normalized-sum fitness plus $10v_B$, and post-hoc repair
Weighted Sum	baseline	one deterministic greedy point ordered by b_j from Section 4.4 (not b_j/c_j and not a front-level weighted-sum solve)
Deterministic Planning	baseline	cost-first greedy fill under budget
Ablation-NoScenarioScreen	ablation	search on all 16 scenarios (screening off)
Ablation-NoRepair	ablation	feasibility repair disabled
Ablation-NoResilienceObj	ablation	survivability column removed from environmental selection; five-objective screening score retained
Ablation-NoOutage	ablation	outage severity zeroed during search; evaluation unchanged

5.2. Protocol

Each stochastic method runs 30 independent optimizer seed indices $r \in \{0, \dots, 29\}$ per experiment label. The archive stores r , while the actual NumPy seed is reconstructed as

$$\text{seed}(m, e, r) = 200000 + 7919r + (\text{int}(\text{SHA1}(\text{p4}|e|m)_{1:6}, 16) \bmod 4096).$$

Because the method name enters the hash, methods do not share paired random-number streams; the Mann–Whitney tests are unpaired. Weighted Sum and Deterministic Planning ignore this seed: the uniform 2640-row archive retains 30 repeated invocations per label, but inference treats each rule as one effective result per label. The seed-level family contains eight stochastic opponents—four stochastic baselines and four ablations—and the sixteen deterministic gaps are descriptive. SHIELD-MOEA, NSGA-II, NSGA-II+Repair, and GA use population 40 and 40 generations. The configured MOEA/D uses 35 reference directions and hence 35 individuals for 40 generations, rather than the nominal population-40 value recorded at the top level of the JSON configuration.

5.3. Metrics and Statistics

The headline metric is standard hypervolume of the budget-feasible raw-objective non-dominated front on the evaluation scenario matrix, computed with pymoo's exact indicator at reference point 1.1 per normalized dimension. Objective bounds are computed once per active evaluation problem from a reference-plan sample generated with NumPy seed 20260713: the empty plan, all 72 single-action plans, and 2048 random plans greedily restricted to the budget, followed by a 5% span margin. No method output sets these bounds. Hypervolume normalization is clipped to $[0, 1]$; search scalarization and compromise selection use the corresponding affine values without clipping. The worst-envelope hypervolume evaluates the mean-front plans using per-objective maxima over the 16 evaluation scenarios and its own fixed bounds. An empty feasible front scores zero.

Statistical comparisons are two-sided Mann–Whitney U tests between SHIELD-MOEA and each of eight stochastic opponents per archive label ($n = 30$ per group), with Holm

correction [44] within each label. Rank-biserial correlation and 5000-resample bootstrap intervals report effect size and mean-difference uncertainty. The deterministic rules receive point comparisons without seed-level p-values. All reported conclusions use the method-independent standard hypervolume defined below.

For reproducibility, the transformations between raw objectives, fronts, statistics, and the AC check are specified explicitly. The final front is first formed in raw objective space:

$$\mathcal{P}_{mer} = \{x \in \mathcal{X}_{mer} : C(x) \leq B, \nexists y \in \mathcal{X}_{mer} \text{ with } F(y) \prec F(x)\}. \quad (413)$$

For objective j , the evaluation reference sample supplies l_j and u_j . Hypervolume then uses

$$\hat{f}_j(x) = \min\left\{1, \max\left[0, \frac{f_j(x) - l_j}{u_j - l_j}\right]\right\}. \quad (416)$$

With reference point $z^{\text{ref}} = (1.1, \dots, 1.1)$, the primary score is the dominated union volume

$$HV(\mathcal{P}_{mer}) = \lambda_d\left(\bigcup_{x \in \mathcal{P}_{mer}} [\hat{F}(x), z^{\text{ref}}]\right), \quad (419)$$

where λ_d denotes d -dimensional Lebesgue measure. The tail-oriented readout replaces each scenario-dependent objective by its observed maximum on the frozen evaluation scenarios,

$$f_j^{\text{wc}}(x) = \max_{s \in \mathcal{S}_{\text{eval}}} f_j(x, s), \quad (423)$$

before applying the same fixed-bound hypervolume construction to the same $\mathcal{P}_{mer}$ plans. The exported compromise is selected from the mean-objective front by an equal-weight affine sum without clipping,

$$x_{mer}^* = \arg \min_{x \in \mathcal{P}_{mer}} \sum_{j=1}^5 \frac{f_j(x) - l_j}{u_j - l_j}. \quad (427)$$

Only the counts of the four action kinds for $r = 0$ are exported to the AC mapper; nodal identities are discarded, and the first minimum is used if the array order contains a tie. For two seed samples of size n_1 and n_2 , the reported rank statistic is

$$U_1 = n_1 n_2 + \frac{n_1(n_1 + 1)}{2} - \sum_{i=1}^{n_1} R_i, \quad (431)$$

and the monotone Holm adjustment over the M within-label comparisons is

$$p_{(i)}^{\text{Holm}} = \max_{k \leq i} \min\{1, (M - k + 1)p_{(k)}\}. \quad (433)$$

Finally, screening workload is reported independently of wall-clock time. Including both active-set union evaluation and full-set ranking rounds, the static code-path reduction relative to full-scenario union evaluation is

$$\eta_{\text{screen}} = 1 - \frac{GN_u K + N_r N_p |\mathcal{S}|}{GN_u |\mathcal{S}|} = 1 - \frac{17,920}{51,200} = 0.65. \quad (437)$$

The separate AC layer declares a mapped case feasible only when all three engineering conditions hold:

Table 3. Pooled leaderboard across eight archive labels. Stochastic methods have 240 runs; deterministic rules summarize eight unique outputs retained as 240 repeated provenance rows.

Method	Role	Mean HV	Std	Worst-case HV	Mean front size	Mean runtime (s)
Ablation- NoResilienceObj	ablation	0.27467	0.02667	0.26976	38.9	0.089
SHIELD- MOEA	proposed	0.27396	0.02700	0.26911	38.9	0.089
Ablation- NoScenarioScreen	ablation	0.27266	0.02627	0.26804	38.9	0.079
Ablation- NoOutage	ablation	0.27125	0.02511	0.26610	38.4	0.087
NSGA-II	baseline	0.25953	0.02816	0.25541	40.0	0.092
NSGA- II+Repair	baseline	0.26070	0.02696	0.25654	40.0	0.091
Ablation- NoRepair	ablation	0.25176	0.02919	0.24649	39.7	0.058
GA	baseline	0.21987	0.02811	0.22107	1.0	0.007
Deterministic Planning	baseline	0.02387	0.00124	0.02282	1.0	0.0004
Weighted Sum	baseline	0.01963	0.00146	0.01805	1.0	0.0004
MOEA/D	baseline	0.00047	0.00002	0.00047	1.0	0.450

Table 4. SHIELD-MOEA vs. NSGA-II. Intervals are pointwise, multiplicity-unadjusted 5000-resample bootstrap CIs for the mean HV difference; r_{rb} is rank-biserial correlation. Holm correction covers eight stochastic opponents per experiment and determines the significance statements.

Experiment	SHIELD	NSGA-II	ΔHV (95% CI)	r_{rb}	Holm p
deterministic_vs_scenario	0.26959	0.25444	0.01515 [0.01096, 0.01945]	0.791	5.89e-7
der_uncertainty	0.27137	0.25471	0.01665 [0.01340, 0.01988]	0.971	6.56e-10
load_uncertainty	0.26211	0.24772	0.01439 [0.01080, 0.01802]	0.842	1.10e-7
outage_contingency	0.26982	0.25224	0.01757 [0.01335, 0.02176]	0.862	5.05e-8
restoration_aware_evaluation	0.33878	0.32708	0.01170 [0.00793, 0.01577]	0.736	5.14e-6
scenario_screening_efficiency	0.26709	0.25481	0.01221 [0.00738, 0.01675]	0.700	1.33e-5
pareto_quality	0.26652	0.25276	0.01375 [0.00882, 0.01868]	0.676	2.89e-5
unseen_stress_generalization	0.24650	0.23247	0.01403 [0.00761, 0.02033]	0.571	7.47e-4

$$I_{AC} = I(\text{converged}) I(0.95 \leq V_b \leq 1.05 \forall b) I(L_\ell \leq 100\% \forall \ell).$$

6. Results

6.1. Main Comparison

Table 3 pools all 8 labels $\times$ 30 seed indices; Figure 2 shows label-level distributions for the six main methods and the pooled mean-versus-worst-envelope panel. This pooled view weights the reference proxy configuration four times because `deterministic_vs_scenario`, `der_uncertainty`, `scenario_screening_efficiency`, and `pareto_quality` differ only in optimizer seed streams under the p4 equations. It is therefore descriptive over archived blocks, not an equally weighted average over eight distinct uncertainty regimes.

SHIELD-MOEA attains pooled mean hypervolume 0.27396 (std 0.02700), 5.09% above NSGA-II with post-hoc final-population repair (0.26070) and 5.56% above plain NSGA-II. Repair explains part of the advantage, but targeted controls do not assign the remainder uniquely to screening or hybrid variation. All 32 comparisons with four stochastic baselines are Holm-significant wins, with no significant loss to any stochastic opponent. Weighted Sum and Deterministic Planning are lower in all sixteen descriptive label-level gaps and receive no seed-level tests.

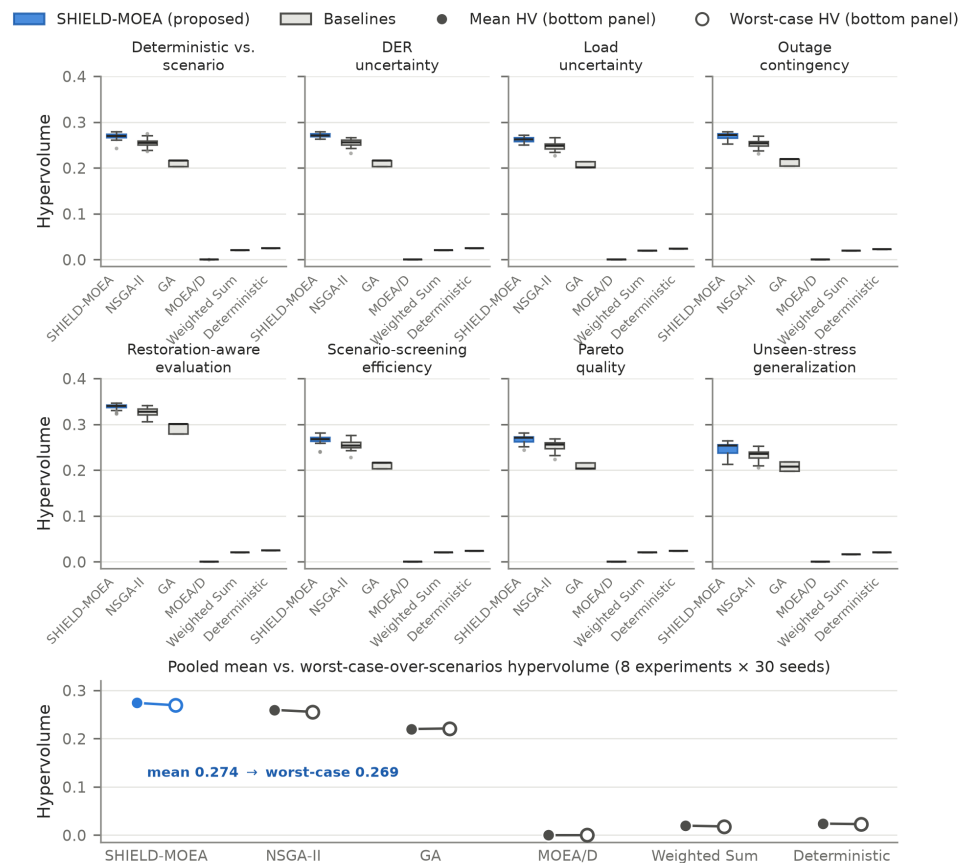


Figure 2. Hypervolume distributions across eight archive labels (30 seeds per box for stochastic methods); deterministic rules are point estimates. The panel text “DER uncertainty” is the preserved archive identifier, not an active-factor claim: δ is absent from p4 objectives. The bottom panel pairs each method’s pooled mean HV (filled) with its sampled worst-envelope HV (open).

The margin is largest in outage_contingency (+6.97%) and in the inactive-factor block labeled der_uncertainty (+6.54%), and remains positive in unseen_stress_generalization (+6.04%). The der_uncertainty margin cannot be attributed to δ because that coordinate is unused by p4 objectives; it reflects another optimizer seed stream on the reference proxy. The unseen-stress label tests changed held-out load and outage ranges, not DER-output shift, and does not certify absence of scenario-distribution specialization.

Three baseline behaviors deserve honest annotation rather than silence. First, MOEA/D collapses on this problem (pooled HV 0.00047): with the budget handled as a penalty, its decomposition weights drive it to near-empty portfolios (mean selected actions 0.004). We report it as configured and note that a constraint-domination variant might fare better; the competitive comparison is carried by NSGA-II. Second, the single-solution methods (GA, Weighted Sum, Deterministic Planning) return one plan by construction, so their hypervolume measures the quality of a point, not a front. Third, runtimes are of the same order for the population methods (SHIELD-MOEA 0.089 s vs. NSGA-II 0.092 s per run at this instance size); no efficiency claim is made from wall-clock at this scale.

6.2. Sampled Worst-Envelope Readout

SHIELD-MOEA’s pooled sampled worst-envelope HV is 0.26911, +5.36% above NSGA-II’s 0.25541 and close to its mean-HV margin. Within the evaluation sample, no mean-envelope reversal is observed against this baseline. This diagnostic describes the 16 sampled scenarios; it is not a bound on unobserved tail behavior.

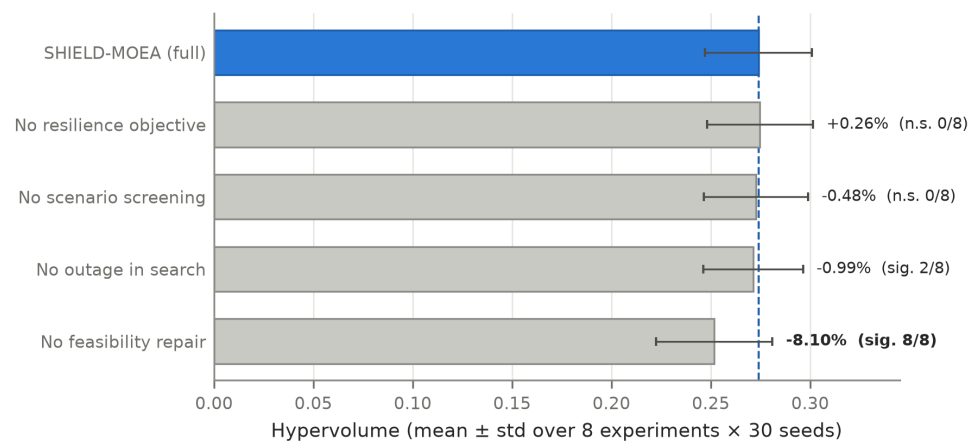


Figure 3. Mean hypervolume (+/- std over 8 archive labels $\times$ 30 seed indices) of SHIELD-MOEA and the four ablations. The plotted legacy label “No resilience objective” means removal of survivability from environmental selection only; screening retains the five-objective scalar. Callouts give the relative difference to the full method and the number of labels (out of 8) in which the difference is Holm-significant.

6.3. Ablation Study: Where the Gain Actually Comes From

Figure 3 compares the full method against the four single-switch ablations, pooled over all experiments and seeds.

The attribution, stated without embellishment:

- SHIELD’s repair schedule is the main contributor.** Disabling it costs 8.10% of pooled hypervolume (0.25176 vs. 0.27396) and is Holm-significant in all 8 labels. Without repair, infeasible plans can occupy slots when too few feasible solutions are available, thinning the feasible population evaluated at the end.
- Outage exposure is associated with a larger AC composition difference in outage-heavy settings.** Under the primary within-label opponent families, zeroing outage severity is significant in `outage_contingency` ($p = 0.032$) and `restoration_aware_evaluation` ($p = 0.009$). Under the supplementary outage-ablation family across eight labels, only `restoration_aware_evaluation` remains significant ($p = 0.024$; `outage_contingency` $p = 0.075$). The fixed-mapping AC layer shows a descriptive composition difference (Section 6.5), not a seeded causal effect.
- Scenario screening has a lower static call count without a resolved front-quality difference.** The screening-off ablation is nominally 0.48% lower and is unresolved in every label. Static code-path arithmetic is 65% lower for screening (Section 4.3), but the implementation does not instrument that count and equivalence was not tested. At this instance size, the full method is slower in the archived runtimes (0.089 s versus 0.079 s per run); a deployment-time saving remains a hypothesis for simulators with expensive scenario evaluation.
- Removing survivability from environmental selection does not raise mean hypervolume.** The ablation yields pooled mean 0.27467, nominally +0.26% above the full method, and is not Holm-significant in any label (0/8; the label-level mean difference ranges from -0.0035 to +0.0039 in absolute HV). This is not a complete no-survivability-search ablation because the five-objective screening score remains active. The supported conclusion is limited to environmental selection: retaining the survivability column exposes a cost-survivability decision coordinate, but no search-quality gain is detected for that use.

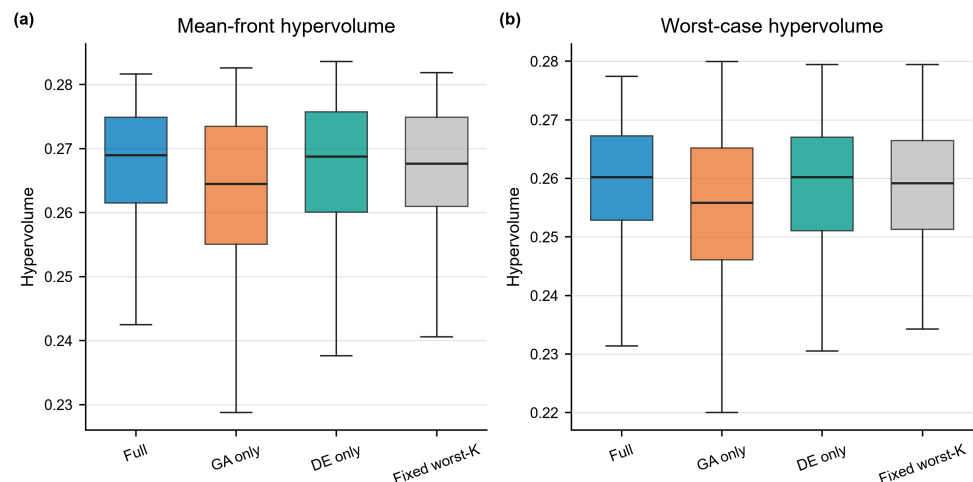


Figure 4. Seed-level mechanism controls across eight archive labels (240 runs per method). (a) Mean-front hypervolume; (b) sampled worst-envelope hypervolume. Boxes pool labels and seeds and are therefore descriptive. Within-label Holm tests show no separation between the full method and DE-only or fixed worst- K in 8/8 labels; the full method exceeds GA-only in 3/8.

Targeted variation and screening controls. Three additional controls use the same eight labels, 30 seed indices, population, generations, search/evaluation split, normalization, and evaluation schedule (720 runs). Pooled mean HV is 0.27425 for DE-only, 0.27408 for a worst- K set fixed after generation 1, 0.27396 for SHIELD-MOEA, and 0.27014 for GA-only. After Holm correction within each label, the full hybrid significantly exceeds GA-only in 3/8 labels and is inseparable in the other five; it is inseparable from DE-only and fixed worst- K in all eight. Thus the data support neither a hybrid-variation gain nor a dynamic-re-screening gain. The narrower interpretation is that the full stack is strong relative to the external baselines, repair is load-bearing, and screening has a lower static objective-row count; DE-only and one-shot worst- K are credible engineering simplifications on this benchmark.

Figure 4 shows the seed-level distributions for those controls. The pooled view is descriptive because label-specific scales differ; the formal verdict remains the within-label Holm analysis. It nevertheless makes both null results visible: the full method overlaps DE-only and fixed worst- K , while GA-only is lower in only part of the archive.

6.4. Parameter Sensitivity Analysis

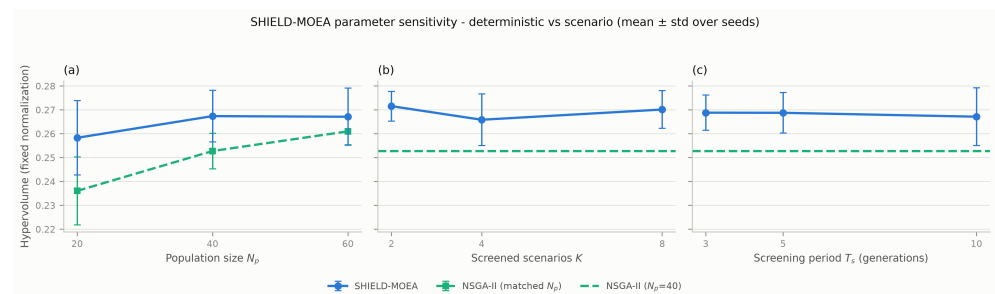
We performed a one-at-a-time sensitivity analysis on the reference experiment (deterministic_vs_scenario). The sweep covers population size $N_p \in \{20, 40, 60\}$ (default 40, with NSGA-II rerun at each matched size), screening width $K \in \{2, 4, 8\}$ (default 4), and screening period $T_s \in \{3, 5, 10\}$ (default 5). Each point uses 10 independent seeds. Figure 5 and Table 5 summarize the results.

Note: the default configuration (population 40, $K=4$, $T_s=5$) appears once per sweep axis, so the table contains three default rows; they show slightly different SHIELD-MOEA means (0.2674, 0.2658, 0.2687) because each axis is drawn from an independent 10-seed stream – a normal consequence of sampling variance, not an inconsistency. For the same reason the NSGA-II reference is identical (0.2527) across the K and T_s rows, which reuse the matched population-40 reference, while the population-size rows re-run NSGA-II at each matched size (0.2360, 0.2527, 0.2610).

SHIELD-MOEA has a higher mean than NSGA-II at all nine parameter points, with a smallest absolute margin of 0.0061. Its mean varies by 5.0% of the default value across the sweep; the K and T_s axes vary by 2.1% and 0.6%, respectively. At $N_p = 60$, the raw

Table 5. Exploratory parameter sweep (10 seeds per point). Reported Mann–Whitney p-values are nominal and multiplicity-unadjusted.

Parameter	Value	SHIELD-MOEA HV (mean +/- std)	NSGA-II reference	p (MWU)
population size N_p	20	0.2583 +/- 0.0155	0.2360	0.011
population size N_p	40 (default)	0.2674 +/- 0.0108	0.2527	0.007
population size N_p	60	0.2671 +/- 0.0120	0.2610	0.104
screening width K	2	0.2715 +/- 0.0062	0.2527	0.0003
screening width K	4 (default)	0.2658 +/- 0.0108	0.2527	0.014
screening width K	8	0.2701 +/- 0.0079	0.2527	0.001
screening period T_s	3	0.2688 +/- 0.0074	0.2527	0.0008
screening period T_s	5 (default)	0.2687 +/- 0.0085	0.2527	0.002
screening period T_s	10	0.2671 +/- 0.0121	0.2527	0.017

**Figure 5.** Parameter sensitivity over population size, screening width K , and screening period T_s (10 seeds per point; NSGA-II re-run at matched population sizes).

comparison is unresolved ($p = 0.104$). Because the sweep has no confirmatory multiplicity family, the section supports only descriptive sensitivity patterns.

6.5. AC Power-Flow Validation on Six Distribution Networks

Hypervolume on proxy objectives cannot certify electrical feasibility. For three selected archive labels, the $r = 0$ equal-sum compromise is reduced to counts of reinforcement, storage, DER, and automation actions and mapped onto four SimBench MV networks (rural, semi-urban, urban, and commercial), the CIGRE MV benchmark, and the IEEE 33-bus feeder. Fixed rules parallel the most-loaded lines, attach storage to the weakest-voltage load buses at $\pm 3\%$ of net load, and add PV equal to 4% of net load at the highest-load buses. Automation has no steady-state effect. The original candidate identities are not transported, the identical mapping logic is used across network families, and no per-network optimizer tuning is performed.

Pandapower [45] solves AC power flow under base, 1.3x peak, 1.5x growth, 1.8x extreme growth, high-DER backfeed, and 1.5x growth with an $N - 1$ outage. The selected planning labels are *deterministic_vs_scenario*, *der_uncertainty*, and *outage_contingency*; the middle label supplies a separate $r = 0$ optimizer composition but is not a DER-aware planning result because δ is inactive in p4. DER-output stress enters here through the high-DER AC operating case. A case is feasible when power flow converges, voltages remain within $[0.95, 1.05]$ pu, and line loading does not exceed 100%.

Each of the twelve planning/reference configurations faces $3 \times 6 \times 6 = 108$ cases, for 1296 AC solves in total. The readout is a qualitative composition-level consistency check, not a seeded replication of the proxy comparison. Table 6 reports the aggregate results; Section 6.7 separates the network families.

Three findings, in decreasing order of importance:

- The mapped compositions improve aggregate feasibility but do not lead the field.** SHIELD-MOEA's compositions are AC-feasible in 68.5% of cases (62.2% under stress-only), against a no-plan reference of 38.9% (33.3% stress-only). Mean minimum

Table 6. AC validation summary over six networks (108 cases per method).

Method	Role	AC-feasible rate	Stress-only feasible	Mean min Vm (pu)	Mean max loading (%)	Mean losses (MW)
NoPlan	reference	0.389	0.333	0.940	78.6	0.569
NSGA-II+Repair	baseline	0.694	0.633	0.969	52.7	0.373
Ablation-NoRepair	ablation	0.694	0.633	0.969	54.3	0.384
SHIELD-MOEA	proposed	0.685	0.622	0.969	52.7	0.377
Ablation-NoResilienceObj	ablation	0.685	0.622	0.970	56.0	0.384
NSGA-II	baseline	0.667	0.611	0.967	52.8	0.376
GA	baseline	0.667	0.600	0.969	48.1	0.362
Ablation-NoScenarioScreen	ablation	0.630	0.567	0.965	51.5	0.390
Ablation-NoOutage	ablation	0.574	0.533	0.969	67.0	0.463
Weighted Sum	baseline	0.417	0.333	0.945	73.7	0.537
Deterministic Planning	baseline	0.417	0.400	0.971	132.0	1.194
MOEA/D	baseline	0.389	0.333	0.940	78.6	0.569

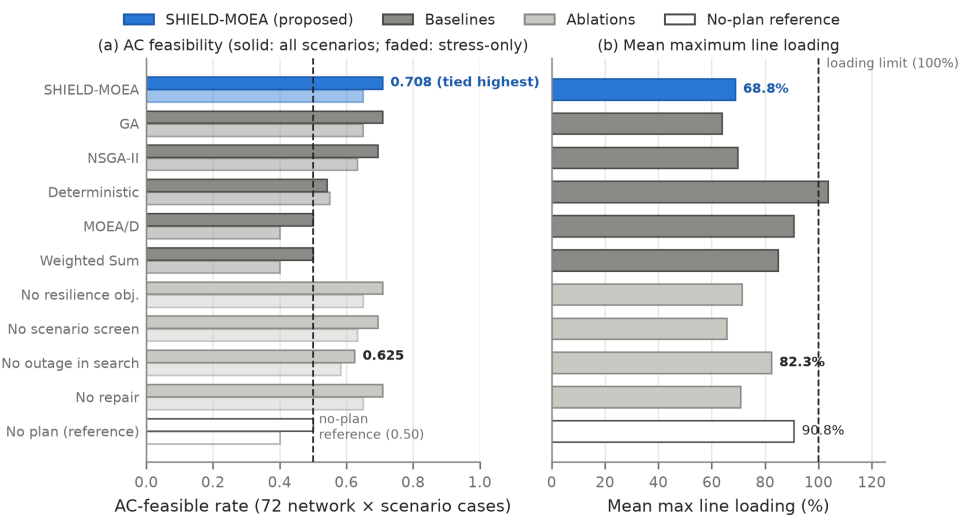


Figure 6. Original SimBench-family AC slice. (a) AC-feasible rate over 72 network-by-scenario cases and over stress scenarios only; the dashed line is the no-plan reference. (b) Mean maximum line loading. The cross-family extension is reported in Table 6 and Figure 9.

- voltage improves from 0.940 to 0.969 pu and mean maximum loading falls from 78.6% to 52.7%. Post-hoc-repaired NSGA-II is slightly higher in feasibility (69.4%), so no AC-superiority claim is made.
- 2. Outage-aware search is the component the physics layer differentiates.** The NoOutage ablation—identical except that search never experiences outage severity—drops to an AC-feasible rate of 0.574 (0.533 stress-only), while mean maximum loading rises to 67.0%. This is composition-level differential evidence, not a seeded causal estimate; it shows that the proxy-level ablation understates a physically relevant change.
 - 3. A cautionary baseline result.** Cost-first Deterministic Planning attains only a 0.417 AC-feasible rate and mean maximum loading 132.0%, because cheap actions add load-side assets without relieving binding corridors. Its loss value is also the largest in Table 6.

6.6. Quality–Cost and Resilience-Composition Diagnostics

The pooled leaderboard alone cannot distinguish a method that is accurate but computationally wasteful from one that obtains a comparable front with less scenario exposure.

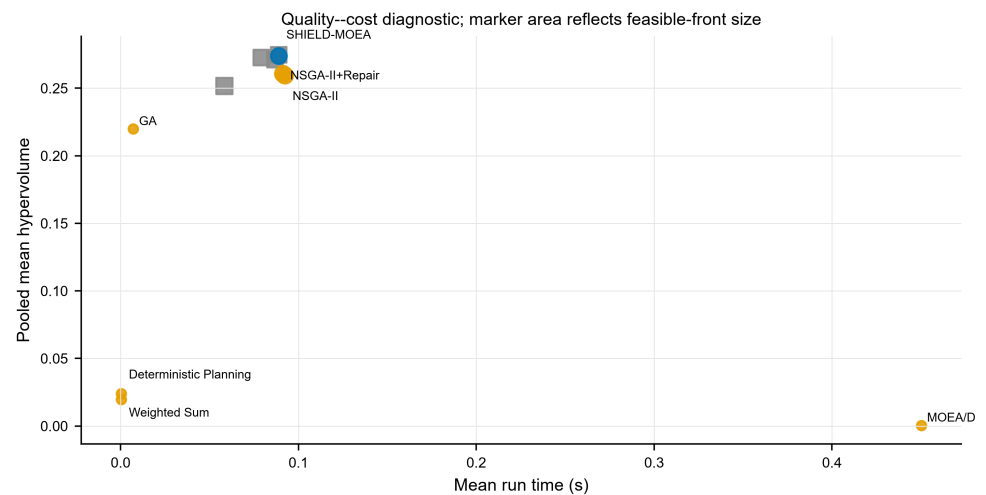


Figure 7. Mean hypervolume versus mean runtime over the eight archive labels and 30 seed indices. Marker area represents mean feasible-front size. The proposed method is highlighted for identification, not to imply statistical dominance over the neighboring ablations.

Table 7. Selected pooled quality–cost readouts from the archived run records.

Method	Mean HV	Mean runtime (s)	Relative HV vs. SHIELD	Interpretation
SHIELD-MOEA	0.27396	0.0889	0.00%	full pipeline
NoScenarioScreen	0.27266	0.0792	-0.48%	no detectable quality loss from screening removal
NSGA-II	0.25953	0.0924	-5.27%	strongest standard evolutionary reference
NoRepair	0.25176	0.0583	-8.10%	repair is load-bearing

Figure 7 therefore places each principal method in the mean-hypervolume/runtime plane using all archived label–seed records. The chart is descriptive: runtimes are sub-second and proxy evaluation is vectorized, so the horizontal coordinate is not a deployment benchmark. It nevertheless shows that SHIELD-MOEA occupies the high-hypervolume region with a runtime comparable to NSGA-II, whereas the configured 35-member penalty-based MOEA/D is both slower and ineffective on this constrained binary problem. The strongest nominal result belongs to the NoResilienceObjective ablation, not to the full method; its difference from SHIELD-MOEA is only 0.26%, and the contrast remains unresolved in all eight labels.

Tables 3 and 7 use the same archived run records, with Table 7 retaining more decimals for selected methods. With $K = 4$ of 16 scenarios, active-set union evaluation is 75% smaller than full-scenario union evaluation. Eight full-set ranking rounds add 5120 objective rows, yielding a static total of 17,920 rows versus 51,200 (65% lower). These are code-path objective-row counts, not instrumented calls or measured energy or time savings.

Figure 8 separates the mean robustness score from its reliability and survivability components. SHIELD-MOEA has pooled reliability 0.9958, survivability 0.9940, and worst-case HV 0.2691. NSGA-II has slightly higher reliability (0.9972) but lower worst-case HV (0.2554); NoScenarioScreen has slightly higher survivability (0.9954) and a similar worst-case HV (0.2680). These crossed rankings are exactly why the paper does not describe the explicit resilience objective or dynamic screening as independently accuracy-improving mechanisms.

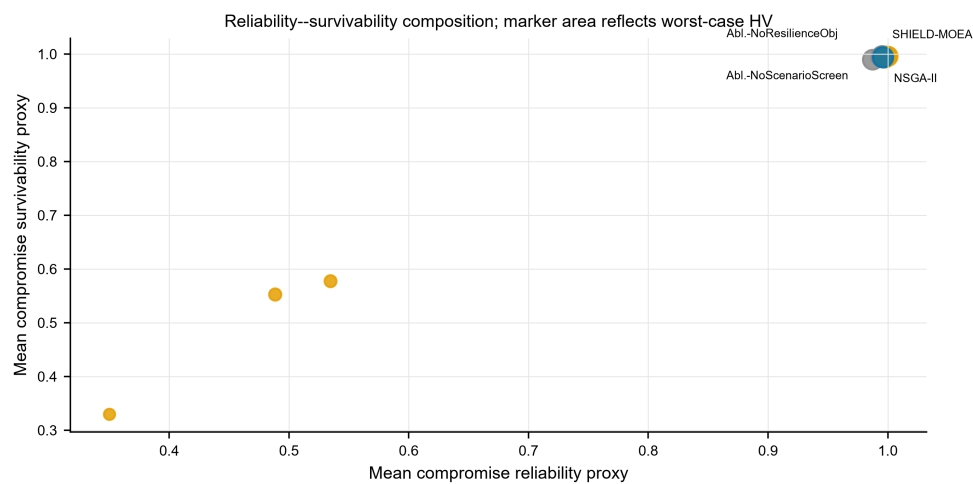


Figure 8. Pooled reliability and survivability of compromise portfolios derived from the frozen evaluation records; marker area represents worst-case hypervolume. The three readouts remain separately encoded rather than being collapsed into a synthetic index.

Table 8. Mechanism-level decision record.

Question	Evaluated comparison	Statistical result	Supported conclusion
Is SHIELD’s repair schedule necessary?	SHIELD vs. NoRepair	significant in 8/8 labels	yes, for front quality in this stack
Is a front-quality difference detected for screening?	SHIELD vs. NoScenarioScreen	no significant difference in 8/8 labels	no difference detected; equivalence not tested
Is periodic re-screening better than a fixed worst-K set?	SHIELD vs. fixed generation-1 worst-K	no significant difference in 8/8 labels	not established
Does hybrid variation beat DE-only?	SHIELD vs. DE-only	no significant difference in 8/8 labels	not established
Does outage exposure matter electrically?	SHIELD vs. NoOutage in AC layer	feasible rate 0.685 vs. 0.574	supported as a composition-level check

Taken together, the diagnostics define the supported mechanism claims. The full pipeline is competitive in proxy quality and has higher sampled worst-envelope HV than the tested stochastic baselines, but the current instance is too cheap for a wall-clock speed claim. Its supported elements are SHIELD’s deterministic repair schedule, lower static scenario-row arithmetic, disjoint-draw scoring, and a descriptive outage-aware composition check. Hybridization and periodic re-screening remain unresolved design choices awaiting a harder simulator-backed benchmark.

6.7. Cross-Family AC Transportability

Figure 9 separates the expanded AC evidence by network family. On the four Sim-Bench networks, SHIELD-MOEA retains the original 0.708 feasible rate. On CIGRE MV it reaches 0.944, compared with 0.167 for no planning, but post-hoc-repaired NSGA-II reaches 1.000. On IEEE 33-bus, both SHIELD-MOEA and post-hoc-repaired NSGA-II reach only 0.333, compared with 0.167 for no planning. The low absolute IEEE rate shows that a composition rule calibrated on SimBench does not automatically transfer to a radial feeder, even when it improves on doing nothing.

7. Discussion

What the evidence supports, component by component. The framework-level result is a 5.56% pooled margin over plain NSGA-II, 32/32 significant stochastic-baseline

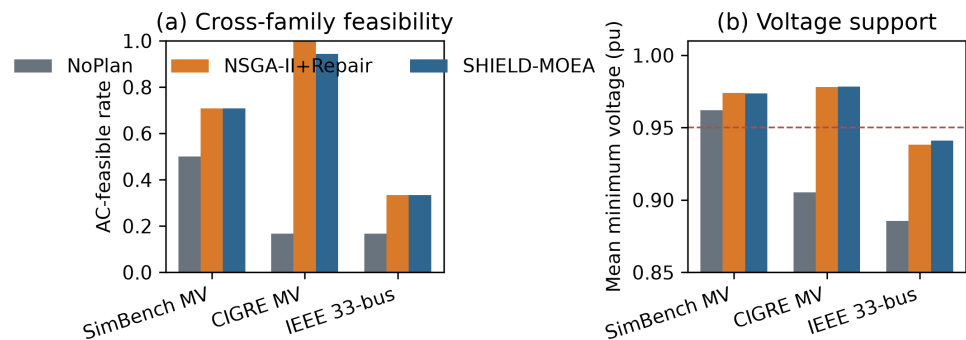


Figure 9. AC-feasible rates and mean minimum bus voltage by network family for the no-plan reference, SHIELD-MOEA, and NSGA-II with post-hoc final-population repair. The dashed voltage line marks the 0.95 pu criterion. Each family uses the same three planning-label compositions and six operating scenarios. SHIELD-MOEA improves over no planning in every family but does not outperform this comparator.

comparisons across eight archived labels, and positive margins in the sampled worst-envelope and held-out-load/outage diagnostics. The pooled result is not an eight-regime DER/load/outage estimate because four labels share the reference p4 configuration. Repair contributes most within SHIELD. Screening has 65% lower static code-path arithmetic while no quality difference is detected against no screening; periodic re-screening does not beat a fixed generation-1 worst- K set, and hybrid GA/DE does not beat DE-only. The AC composition check shows a lower feasibility rate when outage exposure is removed (0.574 versus 0.685), but this fixed-mapping difference is descriptive rather than a seeded causal estimate. The method should therefore be read as an evaluated load/outage-proxy pipeline with repair and direct-reuse leakage control, not as evidence for DER-output robustness or for the necessity of every named operator.

Scenario selection as a design axis. Screening reduces objective calls, and no quality difference is detected against no screening; equivalence is not established. Dynamic updates also do not outperform a fixed generation-1 worst- K set. The supported design principle is selective search evaluation with a disjoint scoring set, not periodic adaptation itself. Using one scenario sample for both search decisions and final grading would leave the reported score vulnerable to scenario overfitting.

Positioning against CARS-MODE. The companion study [42] shares the planning runner, candidate derivation, binary-plan representation, evaluation/statistics utilities, seeded archive machinery, and AC mapping. It self-adapts operator parameters for an economically framed DER-integration problem, while this paper fixes the SHIELD operator parameters and studies scenario exposure for a survivability-framed proxy. The method-specific configurations, archives, comparisons, and conclusions are independent even though the infrastructure is not.

Practitioner reading. The full pipeline produces 4.8–7.0% higher mean hypervolume than post-hoc-repaired NSGA-II across the archived labels, not eight distinct regimes. Ablations associate most of the framework difference with SHIELD's repair schedule rather than screening or hybrid variation. The selected compromise compositions have AC-feasible rates within the range of the strongest alternatives and avoid the loading pattern observed for cost-first selection. Removing outage exposure leaves proxy quality nearly unchanged but reduces AC feasibility in the displayed cases, supporting its inclusion when resilience is the planning objective.

8. Limitations

We state the boundaries of this study explicitly.

1. **DER-output uncertainty is configured but inactive in the p4 proxy.** The multiplier δ enters the hosting diagnostic H , which is omitted from the p4 objective vector. It therefore does not affect p4 search, held-out hypervolume, or compromise choice. The `der_uncertainty` label is a reference-configuration replicate under different optimizer seeds, and the δ shift in `unseen_stress_generalization` is also inactive. Only the AC high-DER case evaluates electrical DER-output stress.
2. **Eight archive labels are not eight distinct proxy regimes.** Four labels share the reference p4 objective configuration; the pooled mean and 32/32 comparison count weight that configuration four times. They summarize the archive and should not be interpreted as broad coverage of eight distinct uncertainty mechanisms.
3. **Recovery and restoration are not modeled.** The scenario coordinate σ scales a survivability proxy, and the planning stage does not model post-event restoration, network reconfiguration, or intentional islanding. Despite its name, `restoration_aware_evaluation` applies L^e in both search and scoring. The $N - 1$ contingency appears only in the AC validation layer. An element-level $N - k$ sampling model with fragility-curve-driven failures and post-event restoration optimization—in the tradition of [10,11]—would deepen the resilience claim and simultaneously decorrelate survivability from reliability (Section 6.3).
4. **Costs are synthetic.** Candidate costs are analytic functions of subnet statistics in an uncalibrated cost unit; no engineering-economic conclusion (payback, budget sizing) should be drawn before calibration against published utility investment data.
5. **Single planning benchmark family.** All optimization labels instantiate one 72-candidate pool from one SimBench complete-mixed extraction. CIGRE MV and IEEE 33-bus extend only the AC validation layer; they do not provide independently optimized candidate pools. The component attribution therefore remains specific to the SimBench-derived planning proxy.
6. **The AC layer validates mapped action-kind counts, not plans or nodal designs.** Only seed-index-0 compromise counts are transported to six networks, and fixed rules reconstruct locations. A nodal siting/sizing formulation on each feeder would provide a stronger transfer test and may change the low IEEE 33-bus feasible rates.
7. **Statistical caveats are part of the record.** The advantage over NSGA-II loses significance at population size 60 in the sensitivity sweep ($p = 0.104$), the scenario-screening ablation remains unresolved in all eight labels, and the NoResilienceObj ablation nominally exceeds the full method. Moreover, NoResilienceObj retains survivability in scenario screening, so it tests environmental selection only.
8. **Proxy objectives and normalization choices bound the claims.** The planning-stage objectives are engineering proxies. Raw objectives define dominance; clipping occurs only inside hypervolume, while search scalarization and compromise choice are unclipped. All electrical statements rest on the pandapower composition layer and do not extend beyond its six networks, six operating cases, and fixed mapping.
9. **The sampled worst-envelope hypervolume is sample-dependent.** It evaluates componentwise maxima over 16 held-out scenarios on the mean-front plans; it neither constructs a worst-objective front nor bounds unobserved tail behavior.
10. **Hybrid and dynamic-screening necessity are not established.** The archived controls return null results: hybrid GA/DE is statistically inseparable from DE-only, and periodic re-screening is inseparable from a fixed generation-1 worst- K set in all eight labels. These mechanisms remain motivated defaults, not demonstrated sources of the headline gain.

11. **The strongest repaired NSGA-II comparator is not timing-matched.** Its repair is applied only after constrained NSGA-II returns its final population, whereas SHIELD repairs initialization and every offspring batch before selection. The 5.09% margin is therefore a framework comparison against post-hoc-repaired NSGA-II, not an isolated repair-timing contrast.

9. Conclusions

This paper asked how an evolutionary resilience-planning proxy can use selective scenario exposure without grading final fronts on the same realizations that guided search. SHIELD-MOEA combines worst- K screening with disjoint search and evaluation draws and reports sampled worst-envelope hypervolume alongside the mean. The archive supports this separation as an implemented evaluation design under active load and outage coordinates; it does not supply DER-output-uncertainty evidence, and periodic updating itself does not improve front quality relative to a fixed worst- K control.

The full method has pooled mean hypervolume 5.09% above post-hoc-repaired NSGA-II and 5.56% above plain NSGA-II over eight archived labels and 30 seed indices per stochastic method. It wins all 32 Holm-corrected comparisons with stochastic baselines; sixteen additional deterministic-rule gaps are descriptive and favor SHIELD-MOEA. Because four labels share the reference proxy configuration, these counts do not represent eight distinct regimes. The method retains a 6.04% mean margin under held-out load/outage ranges and a 5.36% margin on sampled worst-envelope hypervolume. Across four Sim-Bench networks, CIGRE MV, and IEEE 33-bus, its mapped compositions improve aggregate AC feasibility over no planning (0.685 versus 0.389) but trail post-hoc-repaired NSGA-II (0.694).

Component attribution is narrower. SHIELD's initialization-and-offspring repair contributes most within its own ablation. Screening has lower static objective-row arithmetic; no mean-hypervolume difference is detected, and equivalence is not established. Dynamic re-screening does not beat fixed worst- K ; hybrid variation does not beat DE-only; and survivability in environmental selection is a decision-support feature rather than a detected performance feature. Under the fixed composition mapping, removing outage-aware search reduces six-network AC feasibility from 0.685 to 0.574 and raises mean maximum loading from 52.7% to 67.0%.

Post-event recovery, element-level $N - k$ modeling, monetary calibration, a second independently optimized benchmark system, and nodal AC differentiation remain necessary extensions. Within the tested setting, the main contribution is a scenario-selective planning workflow with disjoint final evaluation, a resolved repair effect, and explicit evidence on where screening and hybridization do not add quality.

Author Contributions

[AUTHOR INPUT REQUIRED: insert the final author list, assign CRediT roles, and obtain approval from every author.] All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement

The source networks are public: candidate construction uses SimBench complete mixed network 1-complete_data-mixed-all-0-sw (<https://simbench.de>), and the AC layer uses four SimBench MV networks, CIGRE MV, and IEEE 33-bus through pandapower (<https://www.pandapower.org>). This manuscript package contains the derived manuscript tables and figure scripts. A complete executable supplement containing the benchmark runner, its imported shared hypervolume/non-dominance helper, controlling JSON configurations, 2640 main records, 720 targeted-control records, inference/effect tables, sensitivity records, 1296 AC-case records, and control-run orchestration has not yet been assembled in this package; a persistent release URL/DOI and source-data terms require author confirmation. The deterministic rules are treated as $n = 1$ per label for inference. The shared-infrastructure boundary with CARS-MODE [42] includes the planning runner, source extraction and candidate builder, binary-plan/budget structures, evaluation/statistics utilities, seeded archive machinery, and AC mapping; the research questions, method-specific configurations, run archives, comparisons, and conclusions are paper-specific.

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During the preparation of this manuscript, the authors used an AI-assisted coding environment for software development and a large language model for text editing and readability improvement. The authors reviewed and revised all assisted content and take full responsibility for the intellectual content and conclusions of the publication.

Conflicts of Interest

The authors declare no conflicts of interest.

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